

Heat Loss Analysis for Hotform Furnace

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1 The Plane Wall Method for Heat Loss

While implementing this method to the Hotform furnace, some assumptions were first made. It was assumed that the hot air inside the furnace has a uniform temperature of 900°C everywhere. Looking at the furnace in cross-section, the arrangement was considered as *insulator*, *metal sheet*, and *ambient air* from left to right.

The illustration of the Plane Wall method for this situation can be seen below.



Figure 1: Illustration of Plane Wall Method

In this method, a thermal circuit can be built accordingly.

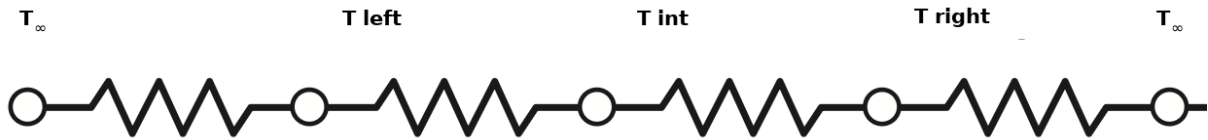


Figure 2: Thermal Circuit

Unknowns

- h_1
- h_2
- $T_{s, \text{left}}$
- T_{int}
- $T_{s, \text{right}}$

Quantity of Interest: The heat flux q'' (heat flow per unit area).

2 Solution Steps

An analytical closed-form solution is not directly attainable because the unknowns are interdependent. We therefore adopt an iterative approach.

Step 1: Initialize q''

Specify an estimated range for the heat flux q'' .

Step 2: Initialize temperature drops

Specify initial guesses for the temperature drops ΔT_1 (between oven gas and the insulator surface) and ΔT_2 (between the metal sheet surface and the ambient air).

Step 3: Update convection coefficients

Using the appropriate convective correlations (laminar assumption in the furnace due to no external forcing), compute h_1 and h_2 from ΔT_1 and ΔT_2 .

$$Ra_L = Gr_L Pr = \frac{g \beta (T_s - T_\infty) L^3}{\nu \alpha} \quad (1)$$

$$Nu_L = 0.68 + \frac{0.670 Ra_L^{1/4}}{\left[1 + (0.492/Pr)^{9/16}\right]^{4/9}} \quad (2)$$

$$Nu_L = \frac{h L}{k} \quad (3)$$

Step 4: Heat flux–temperature consistency

With the updated h values, compute the implied temperature drops via

$$q'' = h \Delta T. \quad (4)$$

Iterate until a consistent combination of q'' and h is found.

Step 5: Temperature distribution across the wall

After ΔT_1 is converged, compute the temperature distribution by integrating

$$\frac{dT}{dx} = -\frac{q''}{k(T)}. \quad (5)$$

This yields a value for $T_{s,\text{right}}$ based on the left-to-right integration through the insulator. Evaluated from the metal side, one obtains another $T_{s,\text{right}}$; a discrepancy indicates mismatch.

Step 6: Bisection method on q''

Define a function $f(q'')$ equal to the difference between the two evaluations of $T_{s,\text{right}}$. The correct solution satisfies $f(q'') = 0$. Using the bracketing range for q'' , evaluate f at the endpoints and at the midpoint, and repeatedly narrow the bracket until the sign changes collapse to zero within tolerance.

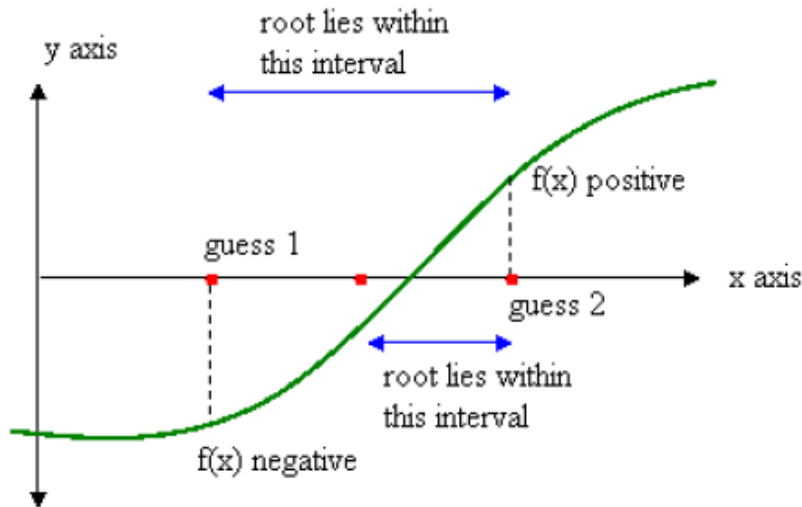


Figure 3: Bisection Method

3 Plane Wall for Upper Surface

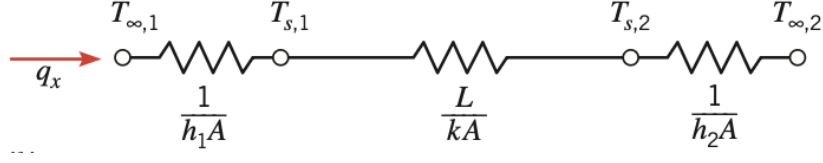


Figure 4: Upper Heat Circuit

For the upper-surface thermal circuit, there is no insulation; thus three regions appear in series. Given h_1 , h_2 , and the metal thermal conductivity k_{metal} and thickness L_{metal} , the resulting heat flux is

$$q''_{\text{top}} = \frac{T_{\infty,1} - T_{\infty,2}}{\frac{1}{h_1} + \frac{L_{\text{metal}}}{k_{\text{metal}}} + \frac{1}{h_2}}. \quad (6)$$

4 Results

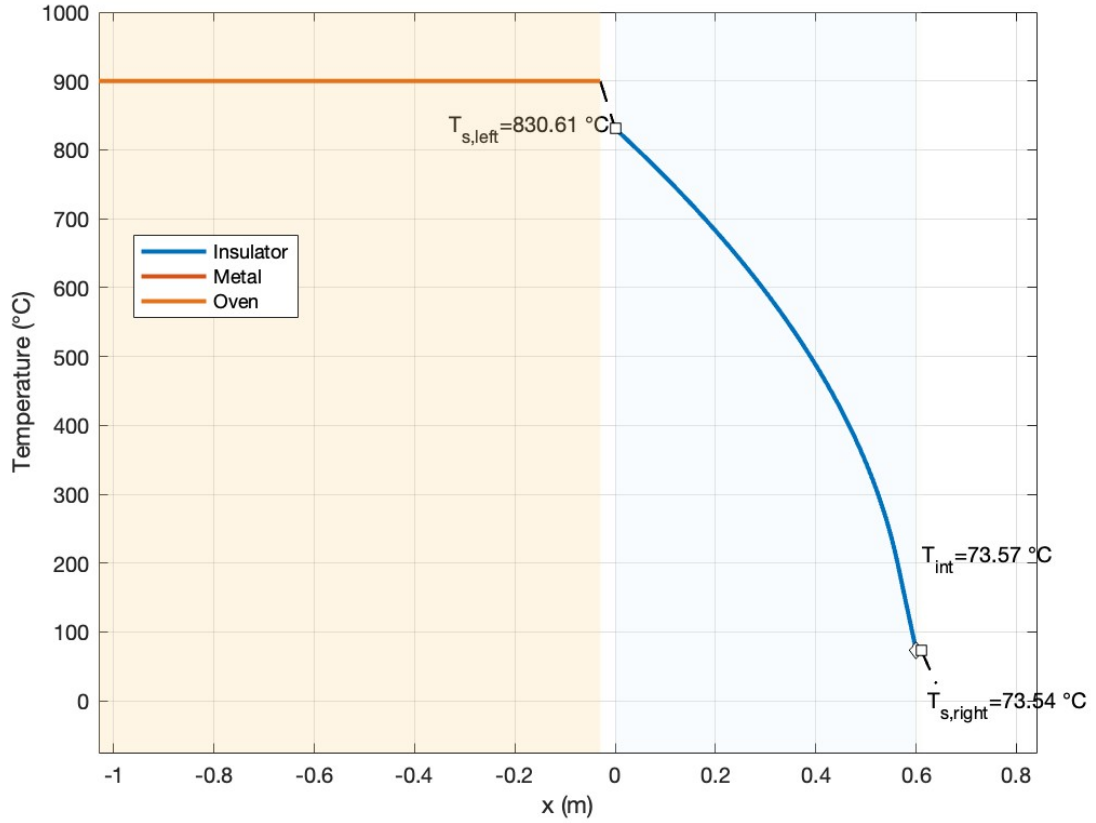


Figure 5: Temperature Graph

$$q'' = 170.961 \text{ W/m}^2, \quad q = 10\,941.522 \text{ W}$$

$$T_{s,\text{left}} = 830.61 \text{ }^\circ\text{C}, \quad T_{\text{int}} = 73.57 \text{ }^\circ\text{C}, \quad T_{s,\text{right}} = 73.54 \text{ }^\circ\text{C}$$

Upper surface:

$$q''_{\text{top}} = 1260.699 \text{ W/m}^2, \quad q_{\text{top}} = 80\,684.758 \text{ W}$$

$$T_{s,\text{top},\text{in}} = 388.28^\circ\text{C}, \quad T_{s,\text{top},\text{out}} = 388.04^\circ\text{C}$$

5 Heat Loss from the Entrance and Exit Gaps

This heat loss occurs due to exchange between hot furnace gases and cold ambient air through the gaps. First, estimate the mass flow rate through these gaps. We consider three different models to gauge sensitivity.

$$Q = C_D A \sqrt{\frac{2 \Delta P}{\rho}} \quad (7)$$

$$Q = 0.21 A_w \sqrt{g H_s \frac{\Delta T}{T_{\text{ref}}}} \quad (8)$$

$$Q = C_o Q_{\text{ideal}} = C_o A_o \sqrt{\frac{2(p_1 - p_2)}{\rho(1 - \beta^4)}} \quad (9)$$

Pressure and density assumptions

Since the furnace is an open system, both internal and external static pressures were taken as atmospheric, $p_0 = 1 \text{ bar}$. Using the ideal-gas relation,

$$\frac{p}{\rho} = RT, \quad (10)$$

the densities inside and outside were computed. Then the hydrostatic pressure difference between inside and outside was estimated as

$$\Delta P = g z (\rho_{\text{out}} - \rho_{\text{in}}). \quad (11)$$

Common constants

$$g = 9.81 \text{ m/s}^2, \quad p_0 = 101\,325 \text{ Pa}, \quad R = 287 \text{ J/(kg} \cdot \text{K)}, \quad c_p = 1007 \text{ J/(kg} \cdot \text{K)}.$$

Model comparison

	Model 1	Model 2	Model 3
Heat loss Q (kW)	105.3	38.37	157.94

Discussion. Model 2 uses a discharge coefficient of $C_d = 0.21$, whereas literature commonly reports $C_d \approx 0.6$ – 0.7 for similar openings. If C_d is adjusted to 0.65 , the result becomes very close to Model 1. In Model 3, the rectangular duct was approximated as a circular pipe via an equivalent diameter.

A MATLAB Codes

Listing 1: HotForm Heat Loss from Side and Upper Surfaces

```

1 Pr = 0.7;   g = 9.81;                               % m/s^2
2
3 B1 = 0.8e-3;   v1 = 162.9e-6;   k1 = 76.3e-3;
4 B2 = 3.2e-3;   v2 = 15.89e-6;   k2 = 26.3e-3;
5
6 L = 1;                               %H
7 A = 2*32*L;                               % m^2
8
9 Lins = 0.6;                               % m
10 Lmetal = 10e-3;                               % m
11 kmetal = 53;                               % W/mK
12
13 Tinf1 = 900;   % C   (left)
14 Tinf2 = 25;   % C   (right)
15
16 m_k = (0.31-0.05)/(1000-200);
17
18 max_iter_outer = 60;
19 tol_q = 1e-6;
20 tol_inner = 1e-8;
21 lambda = 0.7;
22
23 q_lo = 10;
24 q_hi = 500;
25
26 fprintf('Iter | q''''(W/m^2)   dT1(K)   dT2(K)   h1(W/m^2K)   h2(W/m^2K)\n',
27 );
28
29 for it = 1:max_iter_outer
30     qpp = 0.5*(q_lo + q_hi);
31
32     dT1 = 100;
33     for k = 1:80
34         Ra1 = g * B1 * dT1 * L^3 / (v1^2) * Pr;
35         Nu1 = 0.68 + (0.670*Ra1^(1/4)) / (1 + (0.492/Pr)^(9/16))^(4/9);
36         h1 = Nu1 * k1 / L;
37         dT1_new = qpp / max(h1,1e-12);
38         if abs(dT1_new - dT1) < tol_inner, dT1 = dT1_new; break; end
39         dT1 = lambda*dT1_new + (1-lambda)*dT1;
40     end
41     Ts_left = Tinf1 - dT1;
42
43     dT2 = 80;
44     for k = 1:80
45         Ra2 = g * B2 * dT2 * L^3 / (v2^2) * Pr;
46         Nu2 = 0.68 + (0.670*Ra2^(1/4)) / (1 + (0.492/Pr)^(9/16))^(4/9);
47         h2 = Nu2 * k2 / L;
48         dT2_new = qpp / max(h2,1e-12);
49         if abs(dT2_new - dT2) < tol_inner, dT2 = dT2_new; break; end
50         dT2 = lambda*dT2_new + (1-lambda)*dT2;
51     end
52     Ts_right_conv = Tinf2 + dT2;

```

```

53
54 N = 600;
55 dx = Lins / N;
56 T = Ts_left;
57 for i = 1:N
58     if T <= 200
59         k_ins = 0.05;
60     elseif T >= 1000
61         k_ins = 0.31;
62     else
63         k_ins = 0.05 + m_k*(T - 200);
64     end
65     T = T - (qpp / k_ins) * dx;
66 end
67 Tint = T;
68 Ts_right_cond = Tint - qpp * Lmetal / kmetal;
69
70 f_mid = Ts_right_cond - Ts_right_conv;
71
72 fprintf('%3d | %10.3f %7.3f %7.3f %10.3f %10.3f\n', it, qpp,
73         dT1, dT2, h1, h2);
74
75 if it == 1
76     qtmp = q_lo;
77     dT1_lo = 100;
78     for k = 1:80
79         Ra1 = g * B1 * dT1_lo * L^3 / (v1^2) * Pr;
80         Nu1 = 0.68 + (0.670*Ra1^(1/4)) / (1 + (0.492/Pr)^(9/16))
81             ^ (4/9);
82         h1_lo = Nu1 * k1 / L;
83         dT1_new = qtmp / max(h1_lo, 1e-12);
84         if abs(dT1_new - dT1_lo) < tol_inner, dT1_lo = dT1_new;
85             break; end
86         dT1_lo = lambda*dT1_new + (1-lambda)*dT1_lo;
87     end
88     Ts_left_lo = Tinf1 - dT1_lo;
89     dT2_lo = 80;
90     for k = 1:80
91         Ra2 = g * B2 * dT2_lo * L^3 / (v2^2) * Pr;
92         Nu2 = 0.68 + (0.670*Ra2^(1/4)) / (1 + (0.492/Pr)^(9/16))
93             ^ (4/9);
94         h2_lo = Nu2 * k2 / L;
95         dT2_new = qtmp / max(h2_lo, 1e-12);
96         if abs(dT2_new - dT2_lo) < tol_inner, dT2_lo = dT2_new;
97             break; end
98         dT2_lo = lambda*dT2_new + (1-lambda)*dT2_lo;
99     end
100     Ts_right_conv_lo = Tinf2 + dT2_lo;
101     Nlo = N; dxlo = dx; Tlo = Ts_left_lo;
102     for i = 1:Nlo
103         if Tlo <= 200
104             k_ins = 0.05;
105         elseif Tlo >= 1000
106             k_ins = 0.31;
107         else
108             k_ins = 0.05 + m_k*(Tlo - 200);
109         end
110         Tlo = Tlo - (qtmp / k_ins) * dxlo;

```

```

106         end
107         Tint_lo = Tlo;
108         Ts_right_cond_lo = Tint_lo - qtmp * Lmetal / kmetal;
109         f_lo = Ts_right_cond_lo - Ts_right_conv_lo;
110     end
111
112     if f_lo * f_mid <= 0
113         q_hi = qpp;
114     else
115         q_lo = qpp; f_lo = f_mid;
116     end
117
118     if abs(q_hi - q_lo) < tol_q
119         break
120     end
121 end
122
123 qpp = 0.5*(q_lo + q_hi);
124 dT1 = 100;
125 for k = 1:80
126     Ra1 = g * B1 * dT1 * L^3 / (v1^2) * Pr;
127     Nu1 = 0.68 + (0.670*Ra1^(1/4)) / (1 + (0.492/Pr)^(9/16))^(4/9);
128     h1 = Nu1 * k1 / L;
129     dT1_new = qpp / max(h1,1e-12);
130     if abs(dT1_new - dT1) < tol_inner, dT1 = dT1_new; break; end
131     dT1 = lambda*dT1_new + (1-lambda)*dT1;
132 end
133 Ts_left = Tinf1 - dT1;
134
135 dT2 = 80;
136 for k = 1:80
137     Ra2 = g * B2 * dT2 * L^3 / (v2^2) * Pr;
138     Nu2 = 0.68 + (0.670*Ra2^(1/4)) / (1 + (0.492/Pr)^(9/16))^(4/9);
139     h2 = Nu2 * k2 / L;
140     dT2_new = qpp / max(h2,1e-12);
141     if abs(dT2_new - dT2) < tol_inner, dT2 = dT2_new; break; end
142     dT2 = lambda*dT2_new + (1-lambda)*dT2;
143 end
144 Ts_right = Tinf2 + dT2;
145
146 N = 800; dx = Lins/N;
147 x_ins = linspace(0, Lins, N+1);
148 T_prof = zeros(1, N+1); T_prof(1) = Ts_left;
149 for i = 1:N
150     Ti = T_prof(i);
151     if Ti <= 200
152         k_ins = 0.05;
153     elseif Ti >= 1000
154         k_ins = 0.31;
155     else
156         k_ins = 0.05 + m_k*(Ti - 200);
157     end
158     T_prof(i+1) = Ti - (qpp / k_ins) * dx;
159 end
160 Tint = T_prof(end);
161
162 x_met = linspace(Lins, Lins+Lmetal, 30);
163 T_met = Tint - qpp*(x_met - Lins)/kmetal;

```



```

164 Ts_right = T_met(end);
165
166 q = qpp * A;
167
168 fprintf('-----+\n');
169 );
170 fprintf('Bitti (%d iter)\n', it);
171 fprintf('q''' = %.3f W/m^2, q = %.3f W\n', qpp, q);
172 fprintf('T_s,left = %.2f C \n', Ts_left);
173 fprintf('T_int = %.2f C \n', Tint);
174 fprintf('T_s,right = %.2f C \n', Ts_right);
175
176 x0 = 0;
177 xR_bulk = Lins + Lmetal + 0.05*Lins;
178 xL_bulk = -0.05*Lins;
179
180 figure; hold on; box on; grid on;
181
182 % Arka plan bantlar (legend d )
183 patch([0 Lins Lins 0], [Tinf2-100 Tinf2-100 Tinf1+100 Tinf1+100], ...
184       [0.9 0.95 1.0], 'EdgeColor','none','FaceAlpha',0.25,'
185       HandleVisibility','off');
186 patch([Lins Lins+Lmetal Lins+Lmetal Lins], ...
187       [Tinf2-100 Tinf2-100 Tinf1+100 Tinf1+100], ...
188       [0.95 1.0 0.9], 'EdgeColor','none','FaceAlpha',0.25,'
189       HandleVisibility','off');
190
191 % E riler
192 p1 = plot(x_ins, T_prof, 'LineWidth', 2, 'DisplayName',' nslatr ');
193 p2 = plot(x_met, T_met, 'LineWidth', 2, 'DisplayName','Metal');
194
195 % D konveksiyon trend izgileri
196 plot([xL_bulk, 0], [Tinf1, Ts_left], 'k--', 'LineWidth', 1.2);
197 plot([Lins+Lmetal, xR_bulk], [Ts_right, Tinf2], 'k--', 'LineWidth', 1.2)
198 ;
199
200 % Nokta i aretleri
201 plot(0, Ts_left, 'ks', 'MarkerFaceColor','w');
202 plot(Lins, Tint, 'kd', 'MarkerFaceColor','w');
203 plot(Lins+Lmetal, Ts_right, 'ks', 'MarkerFaceColor','w');
204
205 % Etiketler
206 deg = char(176);
207 ofsx = 0.02*Lins;
208 ofsy_left = -0.03*(Tinf1 - Tinf2); % a a kayd r lm
209 ofsy_int = 0.12*(Tinf1 - Tinf2); % yukar
210 ofsy_right = -0.06*(Tinf1 - Tinf2); % a a
211
212 text(0-ofsx, Ts_left+ofsy_left, ...
213      sprintf('T_{s,left}=%.2f %sC', Ts_left, deg), ...
214      'FontSize',10,'HorizontalAlignment','right','VerticalAlignment','
215      bottom');
216 text(Lins+ofsx, Tint+ofsy_int, ...
217      sprintf('T_{int}=%.2f %sC', Tint, deg), ...
218      'FontSize',10,'HorizontalAlignment','left','VerticalAlignment','
219      bottom');
220 text(Lins+Lmetal+ofsx, Ts_right+ofsy_right, ...
221      sprintf('T_{s,right}=%.2f %sC', Ts_right, deg), ...

```

```

216         'FontSize',10,'HorizontalAlignment','left','VerticalAlignment','top
        ');
217
218 xlabel('x (m)');
219 ylabel('Temperature ( C )');
220 axis tight;
221
222 % --- Solda 1 m "Oven" band ve 900 C izgisi ---
223 xL_ambient = xL_bulk - 1.0;
224 ymin = Tinf2 - 100;
225 ymax = Tinf1 + 100;
226 patch([xL_ambient xL_bulk xL_bulk xL_ambient], ...
227       [ymin ymin ymax ymax ], ...
228       [1.0 0.85 0.60], 'EdgeColor','none', ...
229       'FaceAlpha',0.25, 'HandleVisibility','off'); % legend'e girme
230 hOvenLine = plot([xL_ambient xL_bulk], [Tinf1 Tinf1], ...
231                 '-', 'Color',[0.90 0.45 0.10], 'LineWidth',2.0, ...
232                 'DisplayName','Oven');
233 xlim([xL_ambient, xR_bulk+0.2]);
234
235 % Legend (sadece istedi imiz eler )
236 lgd = legend([p1 p2 hOvenLine], {'Insulator','Metal','Oven'}, 'Location',
237             'best');
238
239 % --- st Metal Sac ---
240 R_conv_in = 1/h1;
241 R_cond_met = Lmetal / kmetal;
242 R_conv_out = 1/h2;
243 qpp_top = (Tinf1 - Tinf2) / (R_conv_in + R_cond_met + R_conv_out); % W/m
244 ^2
245 A_top = 32 * 2.0;
246 q_top = qpp_top * A_top;
247 Ts_top_in = Tinf1 - qpp_top * R_conv_in; % C
248 Ts_top_out = Tinf2 + qpp_top * R_conv_out; % C
249
250 fprintf('\n=== st Metal Sac ===\n');
251 fprintf('q''''_top = %.3f W/m^2, q_top = %.3f W\n', qpp_top, q_top);
252 fprintf('T_s,top,in = %.2f C \n', Ts_top_in);
253 fprintf('T_s,top,out = %.2f C \n', Ts_top_out);

```

Listing 2: HotForm Heat Loss from Gaps Model 1

```

1
2 clear; clc;
3
4 x = 2.0;
5 z = 0.10;
6 T_in_C = 900; % C
7 T_out_C = 25; % C
8 n_doors = 2;
9
10 g = 9.81; % m/s^2
11 p0 = 101325; % Pa
12 R = 287; % J/(kg K)
13 cp = 1007; % J/(kg K)
14
15 Aw = x*z;
16 T_in_K = T_in_C + 273.15;

```

```

17 T_out_K = T_out_C + 273.15;
18
19 rho_in = p0/(R*T_in_K);
20 rho_out = p0/(R*T_out_K);
21 rho_ref = 0.5*(rho_in+rho_out);
22 dT = T_in_C - T_out_C;
23
24
25 dP = g * z * (rho_out - rho_in);    % Pa
26
27 C = 0.65;
28
29
30 Qex = C * Aw * sqrt( 2*dP / rho_ref );    % m^3/s
31
32
33 m_out = rho_in * Qex;
34 Q_heat_per_door = m_out * cp * dT;
35 Q_heat_total = n_doors * Q_heat_per_door;
36
37 fprintf('Q_ex = %.4f m^3/s\n', Qex);
38 fprintf('Tek kap      s      kayb = %.1f kW\n', Q_heat_per_door/1000);
39 fprintf('Toplam (%d kap ) = %.1f kW\n', n_doors, Q_heat_total/1000);

```

Listing 3: HotForm Heat Loss from Gaps Model 2

```

1
2 clear; clc;
3
4 x = 2.00;    % m      kap      geni li i
5 z = 0.10;    % m      kap      y ksekli i (=Hs)
6 T_in_C = 900; % C
7 T_out_C = 25; % C
8 n_doors = 2;
9
10 g = 9.81; p0 = 101325; R = 287; cp = 1007;
11
12 Aw = x*z; % m^2
13 Tin = T_in_C + 273.15; % K
14 Tout = T_out_C + 273.15; % K
15 dT = abs(Tin - Tout); % K
16 Tref = 0.5*(Tin + Tout); % K
17 rho_ref = p0/(R*Tref); % kg/m^3
18
19 Vso_m3ps = 0.21 * Aw * sqrt( g * z * (dT / Tref) ); % m^3/s (tek kap
)
20 Vso_Lps = 1000 * Vso_m3ps; % L/s (tek kap
)
21
22 Q_per_door = rho_ref * Vso_m3ps * cp * dT; % W
23 Q_total = n_doors * Q_per_door; % W
24
25 fprintf('Aw=%.3f m^2, Hs=%.3f m, dT=%.1f K, Tref=%.1f K\n', Aw, z, dT,
Tref);
26 fprintf('V_so (tek kap ) = %.5f m^3/s (%.2f L/s)\n', Vso_m3ps, Vso_Lps
);
27 fprintf('Q (tek kap ) = %.0f W (%.2f kW)\n', Q_per_door, Q_per_door
/1000);

```

```

28 fprintf('Q (toplam x%d) = %.0f W (0.2f kW)\n', n_doors, Q_total,
    Q_total/1000);

```

Listing 4: HotForm Heat Loss from Gaps Model 3

```

1
2 T_in_C    = 900;
3 T_out_C    = 25;
4 w_oven     = 2.00;
5 h_oven     = 0.60;
6
7 w_orif     = 2.00;
8 h_orif     = 0.1;
9
10 p0 = 101325;
11 R  = 287;
12 g  = 9.81;
13
14 T_in_K    = T_in_C + 273.15;
15 T_out_K    = T_out_C + 273.15;
16
17 rho_in    = p0/(R*T_in_K);
18 rho_out   = p0/(R*T_out_K);
19
20 dp = g * h_orif * (rho_out - rho_in);
21
22 A_oven = w_oven * h_oven;
23 P_oven = 2*(w_oven + h_oven);
24 D1 = 4*A_oven / P_oven;
25
26 A_orif = w_orif * h_orif;
27 P_orif = 2*(w_orif + h_orif);
28 D2 = 4*A_orif / P_orif;
29
30 beta = D2 / D1;
31
32
33 Co = 0.62;
34 rho = rho_in;
35
36 Q = Co * A_orif * sqrt( 2*dp / ( rho * (1 - beta^4) ) ); % m^3/s
37 v = Q / A_orif; % m/s
38 mdot = rho * Q; % kg/s
39
40
41 cp = 1007; % J/(kg K)
42 dT = T_in_K - T_out_K; % K
43 Qdot_heat = mdot * cp * dT; % W
44 Qdot_heat = 2*Qdot_heat;
45
46 fprintf('Q _heat ( s kayb ) = %.1f W (0.2f kW)\n', Qdot_heat,
    Qdot_heat/1000);

```